

AHMED MUBARAK

647-395-3551 | ahmed.mubarak.eng@gmail.com | [LinkedIn](#) | [Portfolio](#) | Mississauga, ON

SUMMARY

Experience supporting **project planning, schedule and progress tracking, technical documentation, QA/QC, field execution, testing, commissioning, and multidisciplinary project delivery** across infrastructure, industrial, and power-generation projects. Currently at **AECOM**, supporting Project Managers and multidisciplinary teams through engineering design, technical reviews, field activities, FAT/SAT, and commissioning. Previously worked at **GE Vernova as a Project Coordinator / Field Engineer**, supporting Ontario power-station outage execution through scheduling, daily progress tracking, technical documentation, client/vendor coordination, testing, commissioning, and project closeout. with knowledge of **CSA N285/N286**.

***** Holds Bruce Power Nuclear Clearance and Government of Canada Reliability Clearance. *****

EDUCATION

Bachelor of Automation Engineering Technology, (Honours), McMaster University *Sept 2021 - Dec 2025*

- Advanced Diploma, Chemical Engineering Process Design, Mohawk College *Sept 2021 - Dec 2023*
- Project Management Accredited Certificate, McMaster University *Sept 2021 - Dec 2025*

RELEVANT WORK EXPERIENCE

Instrumentation and Controls Junior Engineer - AECOM *Nov 2025 - Present*

- Lead and support assigned engineering project scopes** for major infrastructure projects, coordinating technical requirements, multidisciplinary design inputs, schedules, deliverables, open actions, and execution from design development through **FAT, SAT, and commissioning**.
- Support **Project Managers and discipline leads** with project planning, schedule updates, milestone/progress tracking, scope and deliverable reviews, risk identification, priority management, and follow-through on assigned actions.
- Coordinate with **Engineering, Quality, vendors, contractors, clients, and site teams** to resolve technical issues, track outstanding items, and maintain alignment with **safety, quality, schedule, scope, and project requirements**.
- Coordinate the preparation, review, revision, and tracking of **technical documents and engineering deliverables**, including design packages, drawings, field documentation, comments, revisions, and commissioning records.
- Lead and support **FAT/SAT, commissioning, field verification, deficiency tracking, QA/QC reviews, and technical issue resolution**, ensuring work and deliverables meet applicable project requirements, standards, and quality expectations.
- Perform **I&C design and technical reviews** of P&IDs, instrument lists, schematics, datasheets, and specifications, verifying requirements and coordinating design interfaces across process, electrical, mechanical, vendors, and site teams

Project Coordinator / Field Engineer (Co-op), GE Vernova (General Electric) *Jan 2024 - Dec 2024*

- Supported **Project Managers and field engineering teams** with planning and execution of Ontario power-station outage projects involving gas turbines, steam turbines, generators, excitation systems, and electrical/control maintenance.
- Coordinated **project schedules, daily execution priorities, field progress updates, action items, technical documentation, and activity tracking** to support safe and on-schedule outage execution.
- Supported **inspections, maintenance, repairs, testing, commissioning, issue tracking, documentation review, and closeout**, coordinating engineering requirements with field execution.
- Worked closely with **Project Managers, field engineers, client representatives, vendors, contractors, and site teams** to align scope, schedule, safety, technical requirements, site priorities, and customer expectations.
- Reviewed **engineering drawings, commissioning records, vendor documentation, and field reports** to support troubleshooting, risk identification, progress reporting, and resolution of field challenges.

Automation & Controls Engineer (Co-op), JNE Group

May 2025 – Sept 2025

- Supported **industrial control system design** for US Steel's HDGL wastewater treatment plant by reviewing **P&IDs** and project requirements to develop **PLC panel design, and schematics package**.
- Developed and reviewed **control system deliverables**, including **PLC panel layouts, control schematics**, load lists, field wiring information, installation details & documentation.
- Supported **FAT/SAT preparation**, commissioning readiness, drawing updates, equipment selection, **vendor documentation review**, BOM development, and resolution of technical design comments to support fabrication and field execution.

Control Systems Engineer, Midwest Services Canada

Dec 2024 – May 2025

- Designed, programmed, tested, and commissioned **Siemens PLC-based automation and low-voltage control systems** for commercial and industrial building applications.
- Delivered full design and implementation for the **Delta Marriott Hotel's networked compressor control system**, including **electrical design, PLC programming, and on-site commissioning**.
- **Coordinated directly with clients**, suppliers, and internal team to confirm project requirements, manage design and commissioning priorities, support schedule alignment, and ensure the control system documentation and installation followed applicable **CSA/CEC, ESA/eSAFE**, and project standards.

SKILLS

Project Coordination & Controls: Project planning, schedule coordination, milestone tracking, action logs, meeting records, progress reporting, scope tracking, risk tracking, change tracking, deliverable tracking, closeout support, outage planning, field activity tracking, stakeholder coordination, vendor/client coordination, multidisciplinary team coordination, technical issue resolution, QA/QC support

Engineering & Technical Background: I&C systems, control panels, PLC/SCADA/HMI systems, DCS support, P&ID review, I/O lists, single-line diagrams, loop diagrams, control schematics, electrical drawings, field wiring details, termination details, instrumentation, VFDs, commissioning support, FAT/SAT

Project Management Tools: Microsoft Excel, Microsoft Project, Outlook, SharePoint, Teams, Autodesk Construction Cloud/BIM 360, Primavera P6 exposure, Power BI exposure

Design & Programming Tools: AutoCAD, AutoCAD Electrical, EPLAN, Revit, Studio 5000, RSLogix 5000, Siemens TIA Portal, Siemens LOGO! Soft Comfort, FactoryTalk View, Siemens WinCC, GE Cimplicity/iFIX, Ignition, AVEVA, Nuclear Standards knowledge: CSA N285, CSA N286

ACADEMIC EXPERIENCE

Automation LAB Research Assistant (Part-Time) McMaster University

Sept 2024 – May 2025

- Designing and building an RC car platform with advanced control systems.
- Programmed and wired a conveyor line using **Allen-Bradley PLC** and **HMI** using **Factory Talk View**.
- Configured and tuned (VFDs) for precise motor control in a conveyor system.

Allen Bradley PLC Advanced (3PL3), TA, McMaster University

Sept 2025 - Dec 2025

Allen Bradley PLC (2PL3), TA, McMaster University

Jan 2025 - April 2025

Chemical Engineering Process Design (2EC3), TA, McMaster University

Jan 2025 - April 2025

Math 1 (1MC3), TA, McMaster University

Sept 2024 - Dec 2024

SAMPLE PERSONAL PROJECTS

HVAC Air Compressors Control System – Delta Ottawa Hotel (Capstone)

Jan 2025 – March 2025

- **Led the end-to-end** design and implementation of a dual air compressor building automation system powered by a Siemens LOGO! PLC and CSA-compliant custom panels.
- Designed full system architecture, Electrical schematics, power distribution, control circuits, and protection schemes for fabrication and **ESA/ESAFE** certification.
- Programmed control logic for automatic/manual operation, load balancing, runtime tracking, and maintenance scheduling, extending compressor lifecycle and reliability.
- Integrated advanced alarm handling (leak detection, low pressure, abnormal operation) and automated daily engineer reports, providing operators with real-time visibility of system health.
- **Managed project planning, supplier coordination, commissioning**, and engineer training, delivering a fully certified, standards-compliant solution that reduced downtime and improved operational safety.